Guanyue Qian

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LinkedIn | Google Scholar

Education

New York University *Sept. 2023 – May 2027*

B.S. in Electrical Engineering, GPA: 3.96/4.00

- **Honors and Awards:** William J. Stolze Award (Highest Academic Standing), Myron M. Rosenthal Merit Award, RCA Ted Rappaport Scholarship, Tau Beta Pi, Dean's List (all semesters)

Research Interests

Machine Learning, Wireless Sensing, Computer Vision, Channel Propagation Measurement, Channel Modeling, 3D Reconstruction, Wireless Ray Tracing

Research Experience

HoRAMA: Vision-Language Material-Aware 3D Reconstruction *Feb. 2025 – Present*

Research Advisors: Prof. Theodore S. Rappaport and Prof. David Fouhey

- Implemented **HoRAMA**, a pipeline combining dense SLAM, 3D semantic segmentation, and vision-language material labeling to generate ray-tracing-ready scenes from smartphone video [C5].
- Matched a handcrafted reference model to within 0.1 dB power RMSE at the evaluated factory while reducing scene construction from months of manual work to hours.
- Showed that the required level of geometric detail for wireless digital twin is frequency dependent [C2].

4D Point Splatting: Vision-Guided Differentiable Radar Rendering *Jun. 2026 – Present*

Research Advisor: Prof. Rajalakshmi Nandakumar (Cornell Tech)

- Built a differentiable renderer that synthesizes returns for two 77 GHz, 3T4R MIMO radars from monocular video, fitting scattering parameters and per-joint velocities to measured radar by gradient descent.
- Raised Doppler-centroid correlation from 0.647 to 0.784 on held-out HuPR sequences by anchoring absolute range to the radar and correcting mask-leakage and ground-plane errors in the reconstructed geometry.

NYURay: Machine-Learning Calibration and RIS Optimization *Feb. 2025 – Present*

Research Advisor: Prof. Theodore S. Rappaport

- Calibrated NYURay against dual-band (6.75/16.95 GHz) measurements in an urban microcell scenario and built the evaluation pipeline used for the site-specific validation study [J1].
- Implemented and validated **SPARC**, a sparse ridge-regression calibrator using four ray-path features selected through nested grouped cross-validation, reducing per-path power RMSE from 18.74 to 4.74 dB in a factory and from 23.12 to 5.39 dB in an InH scenario [C3].
- Formulated **PRIMO**, a training-free GPU-parallel optimizer for base-station precoding, RIS phases, and placement, achieving 1.90× the worst-user rate and approximately 100× faster optimization than alternating optimization at 16.95 GHz [C1].

FR1 Radio Propagation Measurement and Channel Modeling *Mar. 2026 – Present*

Research Advisor: Prof. Theodore S. Rappaport

- Conducted FR1 measurement campaigns at 3.7 GHz across outdoor, indoor, and factory environments using a wideband sliding correlator channel sounder.

NYUSIM and IEEE NDAP: Channel Simulation and Data Harmonization *Jun. 2025 – Oct. 2025*

Research Advisors: Prof. Theodore S. Rappaport and Prof. Dan Abraham

- Developed the Python version of **NYUSIM**, from the GUI to core channel-generation algorithms, and verified distributional agreement with v4.0 across the full parameter sweep via Kolmogorov–Smirnov testing [C6].
- Harmonized NYU propagation measurements into a machine-readable point-data schema for IEEE NDAP [C7].

Experience

Teaching Assistant *Jan. 2025 – Dec. 2025*
NYU MakerSpace *Brooklyn, NY*

- Taught fabrication and prototyping workshops (3D printing, laser cutting, water-jet machining) and mentored interdisciplinary student teams from concept to working prototype.

Assistant Baseband Engineer *Jun. 2024 – Aug. 2024*
Quectel Wireless Solutions Co., Ltd. *Shanghai, China*

- Designed schematics and 13-layer PCB layouts with Cadence Allegro for 5G automotive IoT modules.
- Designed automotive module test board (TE-A) for RF/baseband and data transmission tests.

Coach and Robotics Design Lead *Sept. 2020 – Present*
FIRST Robotics Team 8015 & NYU RoboMaster Team *Shanghai & NYC*

- Led robot design, integration, and fabrication for FRC and RoboMaster competitions, spanning CAD, electronics, controls, and software, and coached match strategy to a win at the 2023 Istanbul Regional.

Publications

Journal Articles

[J1] M. Ying, D. Shakya, P. Ma, **G. Qian**, and T. S. Rappaport, "Site-Specific Location Calibration and Validation of Ray-Tracing Simulator NYURay at Upper Mid-Band Frequencies," *npj Wireless Technology*, vol. 2, art. no. 8, Mar. 2026.

Conference Papers

[C1] **G. Qian**, M. Ying, X. Wang, X. Liu, I. S. Gupte, D. Shakya, and T. S. Rappaport, "Joint Beamforming and RIS Deployment Optimization for Outdoor-to-Indoor Coverage at Upper Mid-Band: A Dual-Band Ray-Tracing Study," *Asilomar Conference on Signals, Systems, and Computers*, Pacific Grove, CA, USA, Oct. 2026. *(accepted)*

[C2] **G. Qian**, M. Ying, X. Wang, and T. S. Rappaport, "Impact of Geometric Fidelity on Wireless Channel Prediction Using Site-Specific Ray Tracing in an Indoor Factory at Upper Mid-Band," *submitted to European Conference on Antennas and Propagation (EuCAP)*, Düsseldorf, Germany, 2027.

[C3] M. Ying, **G. Qian**, X. Wang, X. Liu, I. S. Gupte, P. Ma, D. Shakya, and T. S. Rappaport, "SPARC: Sparse Path-Aware Residual Calibration of Wireless Ray Tracing at Upper Mid-Band," *IEEE Global Communications Conference (GLOBECOM)*, Macau S.A.R., China, Dec. 2026. *(accepted)*

[C4] X. Wang, M. Ying, H. Chen, **G. Qian**, X. Liu, P. Ma, D. Shakya, C. Argyropoulos, and T. S. Rappaport, "Robust Hybrid Beamforming with Liquid Crystal Antennas and Liquid Neural Networks," *IEEE 103rd Vehicular Technology Conference (VTC2026-Spring)*, Nice, France, Jun. 2026.

[C5] M. Ying, **G. Qian**, X. Wang, P. Ma, D. Shakya, and T. S. Rappaport, "HoRAMA: Holistic Reconstruction with Automated Material Assignment for Ray Tracing using NYURay," *IEEE International Conference on Communications (ICC)*, Glasgow, UK, Jun. 2026.

[C6] I. Jariwala, X. Wang, B. Meier, **G. Qian**, D. Shakya, M. Ying, H. Nikbakht, D. Abraham, and T. S. Rappaport, "NYUSIM: A Roadmap to AI-Enabled Statistical Channel Modeling and Simulation," *IEEE International Conference on Communications (ICC)*, Glasgow, UK, Jun. 2026.

[C7] D. Shakya, N. A. Abbasi, M. Ying, I. Jariwala, J. Qin, I. Gupte, B. Meier, **G. Qian**, D. Abraham, T. S. Rappaport, and A. F. Molisch, "Standardized Machine-Readable Point-Data Format for Consolidating Wireless Propagation Across Environments, Frequencies, and Institutions," *IEEE Military Communications Conference (MILCOM)*, Los Angeles, USA, Oct. 2025.

Magazine Articles

[M1] L. Belostotski, A. Madanayake, J. Nielsen, M. Abdelhadi, X. Liu, X. Wang, **G. Qian**, S. Ekin, and T. S. Rappaport, "The Anatomy of RF Chains: Metrics, Measures, and Operating Efficiency," *IEEE Microwave Magazine*, 2026. *(accepted)*

Technical Skills

Programming: Python, PyTorch, CUDA, MATLAB, Java, C++, LaTeX

Software: Blender, MeshLab, SolidWorks, Cadence Allegro, Onshape, Fusion 360, Adobe Creative Suite